Devesh Mamidi

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EDUCATION

University of California, Santa Barbara

Bachelor of Science (B.S.) in Computer Science | Santa Barbara, CA

Expected Graduation: December 2028 | GPA: 3.80/4.00 | 2x Dean's Honor List

Relevant Coursework: Problem Solving With Computers, Linear Algebra with Applications, Data Structures and Algorithms, Discrete Mathematics

SOFTWARE ENGINEERING EXPERIENCE

UCSB Caliber

Web Developer | Santa Barbara, CA | Jan 2026 - Present

- Developed a mastery-based course application with assignment, practice question, and instructor analytics workflows, delivering 6+ performance tracking views and summary statistics for a React-based platform used by 300+ students while collaborating with a 10+ member team.
- Resolved up to two assigned issues per sprint in Agile Scrum, implementing 6+ features and 3+ bug fixes for the course platform.

Teach4Speech

Web Developer | Santa Barbara, CA | Jan 2026 - Present

- Developed a difficulty system for the games feature of the app, as measured by having three difficulties for each game, by assigning 1000+ words to each difficulty in the backend and difficulty buttons for each game in the frontend.
- Boosted semantic word-matching accuracy from 66.7% to 95.8% by replacing keyword matching with MiniLM sentence embeddings, building custom Node.js evaluation scripts to benchmark performance across labeled test sets.

TMAS Academy

Web Developer | Remote | July 2025 - Sep 2025; June 2026 - July 2026

- Built a personalized AI chatbot for 500+ STEM AP students with quiz generation and book/problem recommendations to enhance learning and exam preparation.
- Implemented an AI-powered quiz generation feature as measured by dynamically creating up to 10-question assessments by integrating Gemini into the chatbot workflow for STEM AP topics.
- Built a lightweight RAG-powered recommendation system that semantically retrieved the most relevant study materials from 12 PDFs containing 1,000+ practice problems based on each student's selected topic.

SOFTWARE ENGINEERING PROJECTS

SentinelAI

React, TypeScript, Tailwind, FastAPI, ChromaDB, Supabase (PostgreSQL), Docker | Remote | Jul 2026 - Present

- Built an Agentic AI pipeline with RAG (FastAPI, ChromaDB, Gemini Embeddings + Gemini Flash) that performs semantic search over uploaded runbooks, correlates GitHub commit history, and uses prompt-engineered LLM workflows to generate structured root-cause analyses with remediation steps delivered to Slack via webhooks, completing incident analysis in ~8 seconds.
- Developed a full-stack React/TypeScript app with Supabase auth (username/email login, profiles, RLS), project management, and semantic runbook validation (.md/.pdf) by verifying required sections before indexing, with runbooks stored in Supabase.
- Containerized the backend with Docker and a persistent vector store volume for ChromaDB; evaluated system durability trade-offs to retain user-uploaded runbook embeddings across deployments without requiring expensive re-indexing.
- Integrated GitHub, Slack, and Gemini APIs into an async incident orchestration service with structured JSON output and production deployment on Vercel + Render.
- Shipped team collaboration with username/email invites, dashboard notifications, and Owner/Admin/Member RBAC backed by Supabase RLS and PostgreSQL RPCs for invites, role changes, and ownership transfer.
- Built an assign -> fix -> review -> resolve workflow with assignment requests, mandatory fix descriptions, decline-with-feedback resubmissions, and admin auto-resolve paths, enforced end-to-end in Supabase RPCs and surfaced in a role-aware React UI.
- Automated Slack postmortems on incident closure by aggregating AI root-cause analysis, runbook retrieval, GitHub commit evidence, fix submissions, and rejection history into a structured 9-section timeline report via a FastAPI postmortem service.

UCSB Caliber

React, TypeScript, Tailwind, Python, PostgreSQL | Santa Barbara, CA | Jan 2026 - Present

- Developed multiple question formats (multiple choice and free response) as measured by increased assignment and

practice questions flexibility by four times by defining question schemas in PostgreSQL and implementing a React-based frontend.
- Implemented a search feature to search keywords from numerous questions using the searchQuery and searchFilter libraries.
- Implemented an assignments dashboard for teachers and students, enabling creation of multi-question assignments with due dates and student submission tracking by building a React frontend.
- Implemented role-based permissions for teachers and students, enabling teachers to create courses, assignments, and questions and students to join courses and complete assignments and practice questions through a React-based interface.
- Enforced role- and enrollment-based access control across 3 permission levels (admin, instructor, student), gating 9 of 43 authenticated backend endpoints - including grading, grade release, and instructor analytics - to prevent unauthorized access to sensitive academic data.
- Built an instructor analytics dashboard using React/JavaScript (JSX) for interactive table and chart views, JavaScript API client logic for frontend-backend integration, and Python backend aggregation endpoints/schemas to compute and serve course-level student performance metrics (min, mean, median, max, std dev) including at-risk flagging for students with 2+ consecutive submissions below 70%.

TMASChatbot

Next.js, Tailwind, pdf-parse, OpenRouter | Remote | July 2025 - Sep 2025; June 2026 - July 2026

- Developed an AI chatbot supporting 500+ TMAS Academy STEM AP students by integrating Next.js, Tailwind, and OpenRouter models to provide automated study assistance and resource discovery.
- Created up to 10 question interactive quizzes with hints to reinforce mastery of topics using OpenRouter models.
- Built a lightweight RAG pipeline by parsing 12+ PDFs (1,000+ practice questions) with pdf-parse, storing NVIDIA Llama vector embeddings in a JSON-based vector store, and using semantic search to recommend the most relevant study resources.

TECHNICAL SKILLS

Languages: Java, Python, C++, JavaScript, TypeScript, HTML/CSS, SQL
Frameworks and Architectures: Node.js, Tailwind, FastAPI, Flask, REST APIs
Databases: PostgreSQL, Supabase, Firebase, Convex, ChromaDB
Developer Tools: Git, GitHub, Visual Studio Code, Docker, Unix/Linux